최적 제어 hw # 3. 2020311303 윤신혜

1) Problem 4-2

$h(t)$ is continuous $t\ [t_0, t_f]$

$\int_{t_0}^{t_f} h(t)\delta x(t)\,dt = 0$

$\delta x(t)$ is continuous $[t_0, t_f]$ $\delta x(t_0) = \delta x(t_f) = 0$

show $h(t)$: identically zero $[t_0, t_f]$

If $h(t)$ is not zero, positive for $t \in [t_0, t_f]$,

when $t_1, t_2 \in [t_0, t_f]$ and $t_1 < t_2$.

We can select $\delta x(t) = \begin{cases} -(t-t_1)(t-t_2) & \text{for } t \in [t_1, t_2] \\ 0 & \text{The others} \end{cases}$

Then, $\int_{t_0}^{t_f} h(t)\delta x(t)\,dt = \int_{t_1}^{t_2} h(t)\delta x(t)\,dt > 0$

$\therefore$ $h(t)$ is identically zero in $[t_0, t_f]$

Problem
2) 4-5

$$J(x)=\int_{t_0}^{t_f} g(x(t), \dot{x}(t), t)\,dt$$

$x^*+\epsilon\eta$ $\qquad$ $\left.\frac{dJ(x^*+\epsilon\eta)}{d\epsilon}\right|_{\epsilon=0}=0 \Rightarrow$ Euler equation

$$J(x^*+\epsilon\eta)=\int_{t_0}^{t_f} g(x^*(t)+\epsilon\eta(t), \dot{x}^*(t)+\epsilon\dot{\eta}(t), t)\,dt$$

$$\frac{dJ(x^*+\epsilon\eta)}{d\epsilon}=\int_{t_0}^{t_f}\Big[\frac{\partial g}{\partial x}(x^*(t)+\epsilon\eta(t), \dot{x}^*(t)+\epsilon\dot{\eta}(t), t)\,\eta(t)$$

$$+\frac{\partial g}{\partial \dot{x}}(x^*(t)+\epsilon\eta(t), \dot{x}^*(t)+\epsilon\dot{\eta}(t), t)\,\dot{\eta}(t)\Big]dt$$

$$\left.\frac{dJ(x^*+\epsilon\eta)}{d\epsilon}\right|_{\epsilon=0}=\int_{t_0}^{t_f}\Big[\frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t)\,\eta(t)+\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t)\,\dot{\eta}(t)\Big]dt$$

$=0$

$\Rightarrow$ Euler Equation

3) Ex 4-3-3 Find an extremal for the functional

$$J(x) = \int_0^{\pi/4} [x_1^2(t) + \dot{x}_1(t)\dot{x}_2(t) + \dot{x}_2^2(t)]\,dt$$

B.c: $x_1(0) = 1$ $\quad x_1(\frac{\pi}{4}) = 2$

$x_2(0) = \frac{3}{2}$ $\quad x_2(\frac{\pi}{4})$ free

The Euler eq: $\frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t) - \frac{d}{dt}\left[\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t)\right] = 0$

$2x_1^* - \frac{d}{dt}(\dot{x}_2^*) = 0$

$\rightarrow 2x_1^*(t) - \ddot{x}_2^*(t) = 0$ ① $\rightarrow \ddot{x}_2^*(t) = 2x_1^*(t)$

$0 - \frac{d}{dt}(\dot{x}_1^*(t) + 2\dot{x}_2^*(t)) = 0$

$\rightarrow -\ddot{x}_1^*(t) - 2\ddot{x}_2^*(t) = 0$ ② $\rightarrow \ddot{x}_2^*(t) = -\frac{1}{2}\ddot{x}_1^*(t)$

①, ② : $2x_1^*(t) + \frac{1}{2}\ddot{x}_1^*(t) = 0$

$4x_1^*(t) + \ddot{x}_1^*(t) = 0$ $\quad$ Let $x_1^*(t) = ce^{at}$

sol : $x_1^*(t) = c_1\cos 2t + c_2\sin 2t$

$\ddot{x}_2^*(t) = 2x_1^*(t) = 2c_1\cos 2t + 2c_2\sin 2t$

$\dot{x}_2^*(t) = c_1\sin 2t - c_2\cos 2t + c_3$

$x_2^*(t) = -\frac{c_1}{2}\cos 2t - \frac{c_2}{2}\sin 2t + c_3 t + c_4$

Eq 4.3-18 $\delta J(x^*, \delta x) = 0 = \left[\frac{\partial g}{\partial \dot{x}}\right]^T \delta x_f + \left[g - \left(\frac{\partial g}{\partial \dot{x}}\right)^T \dot{x}(t_f)\right]\delta t_f$

$x_1(t_f)$ is specified : $\delta x_{1f} = \delta x_1(t_f) = 0$

$x_2(t_f)$ is free : $\delta x_2(t_f) \neq 0$, $\delta t_f = 0$ (t_f specified)

$\frac{\partial g}{\partial \dot{x}} = 0 = \left[\frac{\partial g}{\partial \dot{x}_2}(x^*(\frac{\pi}{4}), \dot{x}^*(\frac{\pi}{4}))\right] = 0$

$\dot{x}_1^*(\frac{\pi}{4}) + 2\dot{x}_2^*(\frac{\pi}{4}) = -2c_1\sin\frac{\pi}{2} + \cancel{2c_2\cos\frac{\pi}{2}} + 2\left(c_1\sin\frac{\pi}{2} - \cancel{c_2\cos\frac{\pi}{2}}\right) + c_3$

$= -2c_1 + 2c_1 + 2c_{c_3} = 0 \rightarrow c_3 = 0$

$x_1(0) = 1$: $c_1 = 1$

$x_1(\frac{\pi}{4}) = 2$: $c_1 \overset{0}{\cancel{\cos\frac{\pi}{2}}} + c_2 \overset{1}{\cancel{\sin\frac{\pi}{2}}} = 2 \rightarrow c_2 = 2$

$x_2(0) = \frac{3}{2}$: $-\frac{c_1}{2} + c_4 = -\frac{1}{2} + c_4 = \frac{3}{2} \rightarrow c_4 = 2$

$\therefore$ $x_1^*(t) = \cos 2t + 2\sin 2t$

$x_2^*(t) = -\frac{1}{2}\cos 2t - \sin 2t + 2$

Ex 4-4-1 Find a piecewise-smooth curve

$x(0)$, $x(2)=1$, minimize

$$J(x) = \int_0^2 \dot{x}^2(t)[1-\dot{x}(t)]^2\,dt$$

Euler eq : $\frac{\partial g}{\partial x^*} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}^*}\right) = 0$ $\qquad g = \dot{x}^2 - 2\dot{x}^3 + \dot{x}^4$

$$-\frac{d}{dt}(2\dot{x}^* - 6\dot{x}^{*2} + 4\dot{x}^{*3}) = 0$$

$$2\ddot{x}^* - 12\dot{x}^*\ddot{x}^* + 12\dot{x}^{*2}\ddot{x}^* = 2\ddot{x}(\underbrace{1-6\dot{x}+6\dot{x}^2}_{\text{has a real root}})$$

has a real root $\Rightarrow \ddot{x}^*(t)=0$

$$x^*(t) = c_1 t + c_2$$

Weierstrass - Erdmann

i) $\frac{\partial g}{\partial \dot{x}}\Big|_{t=t_1^-} = \frac{\partial g}{\partial \dot{x}}\Big|_{t=t_1^+}$

ii) $g - \frac{\partial g}{\partial \dot{x}}\dot{x}\Big|_{t=t_1^-} = g - \frac{\partial g}{\partial \dot{x}}\dot{x}\Big|_{t=t_1^+}$

i) $2\dot{x}^*(t_1^-) - 6\dot{x}^{*2}(t_1^-) + 4\dot{x}^{*3}(t_1^-)$

$= 2\dot{x}^*(t_1^+) - 6\dot{x}^{*2}(t_1^+) + 4\dot{x}^{*3}(t_1^+)$

ii) $\dot{x}^{*2}(t_1^-)(1-3\dot{x}^*(t_1^-))(1-\dot{x}^*(t_1^-))$

$= \dot{x}^{*2}(t_1^+)(1-3\dot{x}^*(t_1^+))(1-\dot{x}^*(t_1^+))$

인수분해.

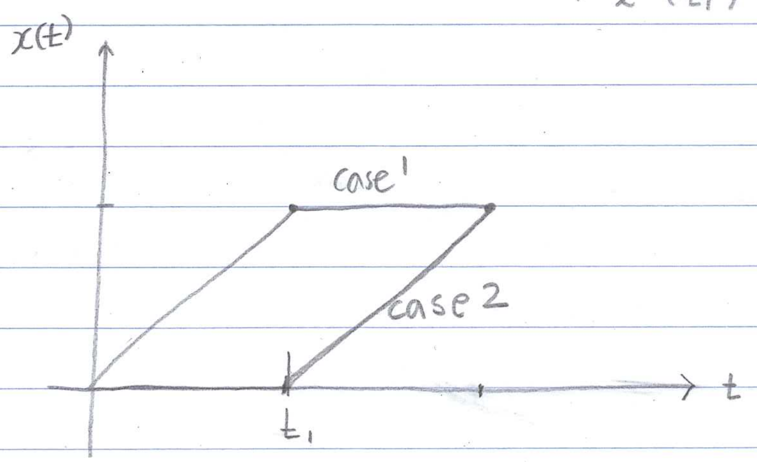
i) $2\dot{x}(1-\dot{x})(1-2\dot{x})$: $\dot{x}(t_1^-) = 0, \frac{1}{2}, 1$

$\dot{x}(t_1^+) = 0, \frac{1}{2}, 1$

ii) $\dot{x}^*(t_1^-) = 0, \frac{1}{3}, 1$

$\dot{x}^*(t_1^+) = 0, \frac{1}{3}, 1$

i), ii) 을 동시에 만족하는 경우 : $\left(\begin{array}{l} \dot{x}^*(t_1^-)=0 \ \& \ \dot{x}^*(t_1^+)=1 \\ \dot{x}^*(t_1^-)=1 \ \& \ \dot{x}^*(t_1^+)=0 \end{array}\right)$ 2가지 경우.



5) Ex 4-5-1 Find the point on the line $y_1+y_2=5$ that is nearest the origin

$$g(y_1, y_2) = y_1^2+y_2^2 \quad \leftarrow \quad y_1+y_2=5$$

$$g(y_1) = y_1^2+(5-y_1)^2$$

$$dg(y_1^*) = \left(\frac{\partial g}{\partial y_1}\right)\Delta y_2 = \{2y_1^* + 2(5-y_1^*)(-1)\}\Delta y_1 = 0$$

$$4y_1^* = 10 \rightarrow y_1^* = 2.5$$

$$\rightarrow g(y_1^*) = \frac{25}{2}$$

$$\therefore (y_1^*, y_2^*) = (2.5, 2.5)$$

6) Ex 4-5-2

$$\begin{cases} y_3 = y_1 y_2 + 5 \\ y_1+y_2+y_3 = 1 \end{cases} \mapsto \text{minimize } f(y_1, y_2, y_3) = y_1^2+y_2^2+y_3^2$$

$$f_a(y_1\ y_2\ y_3\ p_1\ p_2) = y_1^2+y_2^2+y_3^2+p_1(y_1 y_2+5-y_3)+p_2(y_1+y_2+y_3-1)$$

$$\begin{cases} a_i(y_1^*, \dots, y_{n+m}^*) = 0 \\ \frac{\partial f_a}{\partial y_j}(y_1^*, \dots, y_{n+m}^*, p_1^*, \dots, p_n^*) = 0 \end{cases}$$

$$\begin{pmatrix} y_1^*+y_2^*+y_3^*-1=0 \\ y_1^* y_2^* + 5 - y_3^* = 0 \\ 2y_1^* + p_1^* y_2^* + p_2^* = 0. \\ 2y_2^* + p_1^* y_1^* + p_2^* = 0. \end{pmatrix}$$

$$2y_3^* - p_1^* + p_2^* = 0.$$

$$\rightarrow \text{sol}: (y_1^*, y_2^*, y_3^*, p_1^*, p_2^*) = (2, -2, 1, 2, 0)$$
$$= (-2, 2, 1, 2, 0)$$

$$\therefore \text{point}: (y_1^*, y_2^*, y_3^*) = (2, -2, 1)$$

or

$$(-2, 2, 1)$$

7) Ex 4-5-3 N.C (Must be satisfied by the curve of smallest length which lies on the sphere $w_1^2(t)+w_2^2(t)+t^2=R^2$ $(t\in[t_0,t_f])$ specified points w_0, t_0 & w_f, t_f.

minimize $\quad J(w)=\int_{t_0}^{t_f}[1+\dot{w}_1^2(t)+\dot{w}_2^2(t)]^{1/2}\,dt$

$g_a(\vec{w}(t),\dot{\vec{w}}(t),p(t),t)$
$=[1+\dot{w}_1^2(t)+\dot{w}_2^2(t)]^{1/2}+p(t)[w_1^2(t)+w_2^2(t)+t^2-R^2]$

Eq 4.5-30a. $\frac{\partial g_a}{\partial w}(w^*(t),\dot{w}^*(t),p^*(t),t)-\frac{d}{dt}\left[\frac{\partial g_a}{\partial \dot{w}}(w^*(t),\dot{w}^*(t),p^*(t),t)\right]=0$

$2p^*(t)w_1^*(t)-\frac{d}{dt}\left[\dot{w}_1^*(t)[1+\dot{w}_1^{*2}(t)+\dot{w}_2^{*2}(t)]^{-1/2}\right]=0$

$2p^*(t)w_2^*(t)-\frac{d}{dt}\left[\dot{w}_2^*(t)[1+\dot{w}_1^{*2}(t)+\dot{w}_2^{*2}(t)]^{-1/2}\right]=0$

$=0$

$2p^*(t)w_1^*(t)-\left[\ddot{w}_1^*(t)[1+\dot{w}_1^{*2}(t)+\dot{w}_2^{*2}(t)]^{-1/2}-\dot{w}_1^{*2}(t)\ddot{w}_1^*(t)[\quad]^{-3/2}\right]$

$2p^*(t)w_1^*(t)=\ddot{w}_1^*[1+\dot{w}_1^{*2}(t)+\dot{w}_2^{*2}(t)]^{-3/2}\{1+\cancel{\dot{w}_1^{*2}(t)}+\dot{w}_2^{*2}(t)-\cancel{\dot{w}_1^{*2}(t)}\}$

$\therefore$ N.C
$$\begin{cases} 2p^*(t)w_1^*(t)-\ddot{w}_1^*(t)[1+\dot{w}_1^{*2}(t)+\dot{w}_2^{*2}(t)]^{-3/2}[1+\dot{w}_2^{*2}(t)]=0\\ 2p^*(t)w_2^*(t)-\ddot{w}_2^*(t)[1+\dot{w}_1^{*2}(t)+\dot{w}_2^{*2}(t)]^{-3/2}[1+\dot{w}_1^{*2}(t)]=0\\ w_1^{*2}(t)+w_2^{*2}(t)+t^2=R^2\end{cases}$$

8) Ex 4-5-6

$$\dot{x}_1(t) = x_2(t) - x_1(t)$$
$$\dot{x}_2(t) = -2x_1(t) - 3x_2(t) + u(t)$$

~~minimize~~ $J(x, u) = \int_{t_0}^{t_f} \frac{1}{2}[x_1^2(t) + x_2^2(t) + u^2(t)]\,dt$

Let $f_1 = x_2(t) - x_1(t) - \dot{x}_1(t) = 0$

$f_2 = -2x_1(t) - 3x_2(t) + u(t) - \dot{x}_2(t) = 0$

$$g_a(x, \dot{x}, u, p) = \frac{1}{2}x_1^2(t) + \frac{1}{2}x_2^2(t) + \frac{1}{2}u^2(t)$$
$$+p_1(t)[x_2(t) - x_1(t) - \dot{x}_1(t)]$$
$$+p_2(t)[-2x_1(t) - 3x_2(t) + u(t) - \dot{x}_2(t)]$$

Eq 4.5-42a $\frac{\partial g_a}{\partial w}(w^*(t), \dot{w}^*(t), p^*(t), t) - \frac{d}{dt}\left[\frac{\partial g_a}{\partial \dot{w}}(w^*(t), \dot{w}^*(t), p^*(t), t\right] = 0$

$$x_1^*(t) + p_1(t)(-1) + p_2(t)(-2) - \frac{d}{dt}(-p_1^*(t)) = 0$$
$$x_2^*(t) + p_1^*(t)(1) + p_2^*(t)(-3) - \frac{d}{dt}(-p_2^*(t)) = 0$$

N.C
$$\begin{cases} \dot{p}_1^*(t) = -x_1^*(t) + p_1^*(t) + 2p_2^*(t) \\ \dot{p}_2^*(t) = -x_2^*(t) - p_1^*(t) + 3p_2^*(t) \\ u^*(t) + p_2^*(t) = 0 \\ \dot{x}_1^*(t) = x_2^*(t) - x_1^*(t) \\ \dot{x}_2^*(t) = -2x_1^*(t) - 3x_2^*(t) + u^*(t) \end{cases}$$

9) Ex 4-5-7 $\dot{x}_1(t) = -x_1(t) + x_2(t) + u(t)$

$\dot{x}_2(t) = -2x_1(t) - 3x_2(t) + u(t)$

minimize $J(x_0, u) = \int_{t_0}^{t_f} \frac{1}{2}[x_1^2(t) + x_2^2(t)]\,dt$

$\int_{t_0}^{t_f} u^2(t)\,dt = c.$

define $x_1 \triangleq w_1$, $x_2 \triangleq w_2$, $u \triangleq w_3$

Let $u^2(t) = w_3^2(t) = \dot{z}(t)$, $z(t_0) = 0$, $z(t_f) = c$

$$g_a(\vec{w}(t), \dot{\vec{w}}(t), p(t), \dot{z}(t)) = \frac{1}{2}w_1^2(t) + \frac{1}{2}w_2^2(t)$$
$$+ p_1(t)[-w_1(t) + w_2(t) + w_3(t) - \dot{w}_1(t)]$$
$$+ p_2(t)[-2w_1(t) - 3w_2(t) + w_3(t) - \dot{w}_2(t)]$$
$$+ p_3(t)[w_3^2(t) - \dot{z}(t)].$$

Eq 4.5-42 a $\frac{\partial g_a}{\partial w}(w^*(t), \dot{w}^*(t), p^*(t), t) - \frac{d}{dt}\left[\frac{\partial g_a}{\partial \dot{w}}(w^*(t), \dot{w}^*(t), p^*(t), t\right] = 0$

$$\begin{cases} w_1^*(t) - p_1^*(t) - 2p_2^*(t) - \dot{p}_1^*(t) = 0. \\ w_2^*(t) + p_1^*(t) - 3p_2^*(t) - \dot{p}_2^*(t) = 0 \\ p_1^*(t) + p_2^*(t) + 2p_3^*(t)\,w_3^*(t) = 0 \\ 0 - \dot{p}_3^*(t) = 0. \\ \dot{w}_1^*(t) = -w_1^*(t) + w_2^*(t) + w_3^*(t). \\ \dot{w}_2^*(t) = -2w_1^*(t) - 3w_2^*(t) + w_3^*(t) \\ \dot{z}^*(t) = w_3^{*2}(t) \quad z^*(t_0) = 0 \quad z^*(t_f) = c \end{cases}$$